Aviona: Cost to Bring to Market

Initial and recurring costs, including certification | PS2 Automated TCAS for General Aviation

These are order-of-magnitude estimates, not quotes. The biggest variable is **how much the system is allowed to do**. A box that only warns can skip most certification; a box that commands the autopilot needs the full process. So the realistic plan is three phases, each funding the next.

The three phases

Phase	What it does	Approval path	Initial cost	Time
1. Portable advisory unit	Warns and negotiates over LoRa; the pilot flies the turn	None for the aircraft (carried like a portable ADS-B receiver); FCC Part 15 for the radio	~$1–2M	12–18 months
2. Installed traffic advisory	Same, permanently installed, wired to the audio panel and pilot displays	FAA NORSEE policy, plus installation approval	+ ~$1–3M	+1–2 years
3. Full Aviona: automatic avoidance	Commands the autopilot after the countdown	Supplemental Type Certificate with certified software and hardware	+ ~$5–15M	+2–4 years

Phase 1 is mostly engineering:

- A custom circuit board (ESP32 + LoRa + GNSS + barometer + audio), firmware rewritten in C++, an enclosure, FCC testing, and flight testing.
- A team of about 4 engineers for 12–18 months at roughly $150–200k per engineer per year is ~$0.75–1.2M.
- Plus FCC testing (~$15–40k), flight tests in rented Cessnas (~$150–200/hour, so ~$40k for 100 hours each on two aircraft), tooling, legal, and insurance.

Phase 2 uses a cheaper FAA route built for exactly this kind of product. NORSEE covers equipment like traffic advisory systems where an equipment failure would only have a minor effect. One avionics company said it cut their cost of bringing products to market by about 90%. The catch is that NORSEE only makes the equipment eligible for installation; installing it still needs its own approval, such as a field approval or a Supplemental Type Certificate. Budget for:

- environmental and interference testing (the DO-160 standard),
- FAA documentation,
- and an installation approval covering many aircraft models at once.

Phase 3 is where costs jump. Software that can steer the aircraft has failure effects of “major” or worse, so it must meet DO-178C, the FAA’s standard for flight software. Typical DO-178 certification runs **$25 to $100 per line of code**. A certified embedded version of Aviona might be 15,000–30,000 lines, so **$0.4–3M for software alone**. Add the system safety analysis, full environmental testing, integration with each autopilot model (ideally with the autopilot maker as a partner), and the FAA project itself.

Total to reach full automatic avoidance: roughly $7–20M over 4–7 years. That's why the Honeywell angle matters: they already have certified autopilots, certification teams, and FAA relationships, which removes the most expensive and slowest part.

Recurring costs

Per unit:

Item	Phase 1 (portable)	Phase 3 (installed, certified)
Hardware + assembly + test	~$60–120	~$300–800 (aviation-grade parts, connectors, production quality inspections)
Installation	$0	~$1,000–2,500 (5–15 shop hours, antenna, wiring, paperwork)
Owner costs per year	~$0 (battery, optional app features)	~$0–100 (no mandatory fees)
Likely selling price	~$400–800	~$3,000–8,000 + installation

Even the certified version is a fraction of TCAS's $25,000–100,000+.

Per year, for the company:

Item	Rough annual cost
Aviation product liability insurance	~$50–250k
Quality system and production certification audits (installed product)	~$50–150k
Re-certifying software changes	~$50–200k per significant update
Monitoring reported problems and FAA follow-up	staff time
Customer support and firmware updates	1–3 staff
Parts obsolescence (the ESP32 is a consumer chip, not built for 20+ years of supply)	stockpiling parts ahead of discontinuation, or redesigning around new ones

Is it worth it?

Working from the middle of these ranges:

- **Upfront cost:** about $10M through Phase 3.
- **Profit per unit:** about $2,000.
- **Break-even:** about 5,000 aircraft, roughly 2% of the ~214,000 active US general aviation fleet in the business slide.
- **Earlier revenue:** Phase 1 units start selling years before the certified version exists.

How to say it in the pitch

> *"We'd go to market in three steps. First, a portable advisory unit for about $1–2M of development, selling for a few hundred dollars. Second, an installed version through the FAA's NORSEE path for safety equipment. Third, autopilot-coupled avoidance through a Supplemental Type Certificate, roughly $7–20M in total. With Honeywell's certified autopilots and certification experience, the third step becomes integration rather than invention."*

Give these as rough estimates; judges respect a realistic range more than a precise number you can't defend.

Sources

- DO-178 certification at $25–100 per line of code: Synopsys/Coverity white paper "Risk Mitigation for DO-178C".
- NORSEE scope (traffic advisory systems, minor failure condition; eligibility, not installation approval): FAA press release, July 2016, as reported by Skies Magazine and AVweb.
- NORSEE cutting development and market-entry cost by about 90% (AeroVonics): Runway Girl Network, July 2019.
- All other figures are engineering estimates for planning, not vendor quotes.